

Tutorial 3

Three counts (numbers of individuals per 10 m²) in four seasons for two different species of stick insect were determined with the results in Table 1.21. Analyse and interpret these data.

Season	Megacrania sp	Extatosoma sp	Total
Spring	15	13	
	18 (55)	12 (36)	
	22	11	91
Summer	12	15	
	14 (42)	19 (55)	
	16	21	97
Autumn	10	23	
	11 (30)	25 (74)	
	9	26	104
Winter	6	30	
	8 (21)	29 (90)	
	7	31	111
Total	148	255	403

Note: Numbers in parentheses are subtotals for each season and species.

1. What design has been used here and what is the treatment structure? Be specific in naming the design.

This is CRD and the treatment structure is 2 by 4 Factorial.

2. Why is this not a randomized complete block design?

If this was a complete blocked design and say Season was the blocking variable, then we would only have one replicate of treatments (in this case, megacrania and extatosoma) per block. This is also true if species was the blocking variable.

3. Identify the following based on the experiment description:

- a. Treatment factor(s): Season and Species
- b. Factor levels: Season: Spring, summer, autumn and winter. Species: Megacrania, Extatosoma.
- c. Treatments: Spring-Megacrania, Summer-Megacrania, Autumn-Megacrania, Winter-Megacrania, Spring-Extatosoma, Summer- Extatosoma, Autumn-Extatosoma, Winter- Extatosoma.
- d. Response: Counts

- e. Experimental units: an individual count (the action of going out and counting) and there are $r = 3$ times and with 8 treatments, that means $3 \times 8 = 24$ experimental units.
 - f. Number of replicates: 3
4. How was randomization done?

Randomisation was done without restriction.

5. Write down the population model equation used for analysing data from this type of experiment. Be specific.

$$y_{ijk} = \mu + A_i + B_j + (AB)_{ij} + \varepsilon_{ijk}$$

where:

- y_{ijk} = response variable (count) where $i = 1, 2$ and $j = 1, \dots, 4$ and $k = 1, 2, 3$
- μ = overall mean count
- A_i = main effect for species
- B_j = main effects for season
- $(AB)_{ij}$ = interaction effect of i th level of species and j th level of season
- ε_{ijk} = random error term and $\varepsilon_{ijk} \sim N(0, \sigma^2)$

6. What are the assumptions of this model?

1. Homogeneity of variances
2. Normality of errors
3. Independent errors
4. No outliers

7. Calculate (without using R) and define the following:

- a. The treatment means.

$$\overline{Y}_{\text{Spring, Megacrania}} = \frac{15 + 18 + 22}{3} = \frac{55}{3} = 18.333 \dots$$

$$\overline{Y}_{\text{Spring, Extatosoma}} = \frac{13 + 12 + 11}{3} = \frac{36}{3} = 12.000$$

$$\overline{Y}_{\text{Summer, Megacrania}} = \frac{12 + 14 + 16}{3} = \frac{42}{3} = 14.000$$

$$\overline{Y}_{\text{Summer, Extatosoma}} = \frac{15 + 19 + 21}{3} = \frac{55}{3} = 18.333 \dots$$

$$\overline{Y}_{\text{Autumn, Megacrania}} = \frac{10 + 11 + 9}{3} = \frac{30}{3} = 10.000$$

$$\overline{Y}_{\text{Autumn, Extatosoma}} = \frac{23 + 25 + 26}{3} = \frac{74}{3} = 24.666 \dots$$

$$\overline{Y}_{\text{Winter, Megacrania}} = \frac{6 + 8 + 7}{3} = \frac{21}{3} = 7.000$$

$$\overline{Y}_{\text{Winter, Extatosoma}} = \frac{30 + 29 + 31}{3} = \frac{90}{3} = 30.000$$

These are the mean count for each treatment. For example, for summer and extatosoma, the mean count of individuals was roughly 19. However, in winter the mean count of extatosoma was 30.

b. Overall mean

$$\overline{Y}_{..} = \frac{\sum_{i=1}^2 \sum_{j=1}^4 \sum_{k=1}^3 Y_{ijk}}{24} = \frac{403}{24} = 16.791666 \dots$$

or the mean of the treatment means.

This is the average count irrespective of species or season.

c. Main effects

Species

$$\overline{Y}_{\text{Megacrania}} = \frac{55 + 42 + 30 + 21}{12} = \frac{148}{12} = 12.333 \dots$$

$$\overline{Y}_{\text{Extatosoma}} = \frac{36 + 55 + 74 + 90}{12} = \frac{255}{12} = 21.250$$

$$\text{Main effect (Megacrania)} = \overline{Y}_{\text{Megacrania}} - \overline{Y}_{..} = 12.333 \dots - 16.791 \dots = -4.458 \dots$$

$$\text{Main effect (Extatosoma)} = \overline{Y}_{\text{Extatosoma}} - \overline{Y}_{..} = 21.250 - 16.791 \dots = 4.458 \dots$$

These main effects are the average change in counts ignoring season relative to the overall mean. So, the main effect of Megacrania says that on average the mean count of individuals was roughly 5 individuals less than the overall average. On the other hand, the mean count of Extatosoma individuals was higher than average, ignoring season.

Season

$$\overline{Y}_{\text{Spring}} = \frac{55 + 36}{6} = \frac{91}{6} = 15.166 \dots$$

$$\overline{Y}_{\text{Summer}} = \frac{42 + 55}{6} = \frac{97}{6} = 16.166 \dots$$

$$\overline{Y}_{\text{Autumn}} = \frac{30 + 74}{6} = \frac{104}{6} = 17.333 \dots$$

$$\overline{Y}_{\text{Winter}} = \frac{21 + 90}{6} = \frac{111}{6} = 18.500$$

Main effect (Spring) = $15.166\dots - 16.791\dots = -1.625\dots$

Main effect (Summer) = $16.166\dots - 16.791\dots = -0.625\dots$

Main effect (Autumn) = $17.333\dots - 16.791\dots = 0.541\dots$

Main effect (Winter) = $18.500 - 16.791\dots = 1.708\dots$

Mean count of insects (so ignoring the different species) was slightly higher than average in autumn and winter, and slightly lower than average in summer and spring!

d. Interaction effects

The interaction effect is calculated as:

$$(AB)_{ij} = \overline{Y}_{ij} - \left(\overline{Y}_{..} + (\overline{Y}_{i.} - \overline{Y}_{..}) + (\overline{Y}_{.j} - \overline{Y}_{..}) \right)$$

Which simplifies to:

$$(AB) = \overline{Y}_{ij} - \overline{Y}_{i.} - \overline{Y}_{.j} + \overline{Y}_{..}$$

We compute each:

$$\begin{aligned} (AB)_{\text{Spring, Megacrania}} &= 18.333\bar{3} - 12.333\bar{3} - 15.166\bar{6} + 16.791666 \dots \\ &= 7.625000 \dots \end{aligned}$$

$$\begin{aligned} (AB)_{\text{Spring, Extatosoma}} &= 12.000 - 21.250 - 15.166\bar{6} + 16.791666 \dots \\ &= -7.625000 \dots \end{aligned}$$

$$\begin{aligned} (AB)_{\text{Summer, Megacrania}} &= 14.000 - 12.333\bar{3} - 16.166\bar{6} + 16.791666 \dots \\ &= 2.291666 \dots \end{aligned}$$

$$\begin{aligned} (AB)_{\text{Summer, Extatosoma}} &= 18.333\bar{3} - 21.250 - 16.166\bar{6} + 16.791666 \dots \\ &= -2.291666 \dots \end{aligned}$$

$$\begin{aligned} (AB)_{\text{Autumn, Megacrania}} &= 10.000 - 12.333\bar{3} - 17.333\bar{3} + 16.791666 \dots \\ &= -2.875000 \dots \end{aligned}$$

$$\begin{aligned} (AB)_{\text{Autumn, Extatosoma}} &= 24.666\bar{6} - 21.250 - 17.333\bar{3} + 16.791666 \dots \\ &= 2.875000 \dots \end{aligned}$$

$$\begin{aligned} (AB)_{\text{Winter, Megacrania}} &= 7.000 - 12.333\bar{3} - 18.500 + 16.791666 \dots \\ &= -7.041666 \dots \end{aligned}$$

$$(AB)_{\text{Winter, Extatosoma}} = 30.000 - 21.250 - 18.500 + 16.791666 \dots = 7.041666 \dots$$

$(AB)_{\text{Winter, Extatosoma}}$: For winter and extatosoma, the average count of individuals was higher than expected if the season and species did not interact.

$(AB)_{\text{Summer, Megacrania}}$ =: In summer, the mean count of megacrania insects, was higher than if season and species did not interact.

8. Write out the model equation for observations belonging to one of the treatments (you can choose which treatment). Use the estimates rounded to two.

$$Y_{\text{spring,extatosoma},k} = 16.79 + 4.46 - 1.63 - 7.63 + e_{\text{spring,extatosoma},k}$$

9. What is the formula for calculating standard error for treatment means? For main effects? Where would you find or calculate the necessary values?

$$\text{Var}(\hat{u}_{ij}) = \frac{\sigma^2}{n}$$

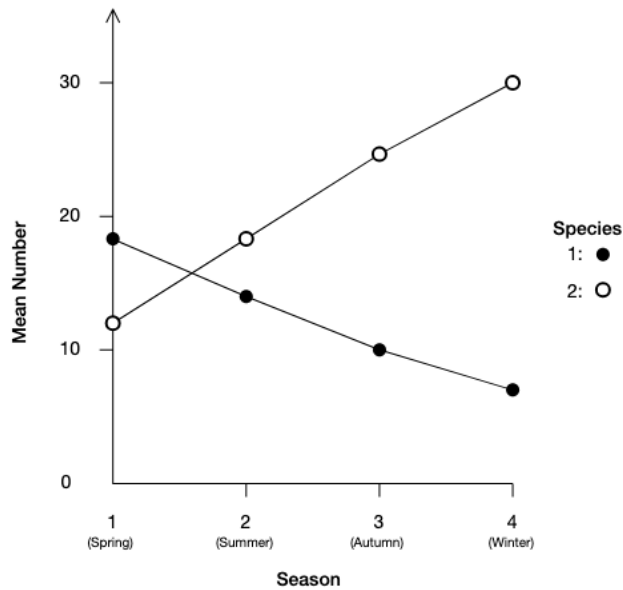
$$\text{Standard error} = \sqrt{\left(\frac{\sigma^2}{n}\right)}$$

The MSE is an estimate of σ^2 and can be found by calculating the error sums of squares and dividing by the appropriate degrees of freedom:

$$\text{MSE} = \frac{\sum(Y_{ijk} - Y_{ij.})^2}{ab(n-1)}$$

These values can be extracted from the ANOVA table.

10. Construct an interaction plot and identify what effects are present.



For Megacrania, the numbers of individuals decreases from Spring to Winter, but for Extatosoma numbers of individuals increases from Spring to Winter.

There is a greater number of Extatosoma than Megacrania.

Note: when a significant interaction is present (as in this example), the interaction effect is usually of much greater interest than the effects of Factor A and Factor B, and the latter should be interpreted with caution.

11. What are the hypotheses for this ANOVA test?

For this ANOVA, we have three sets of hypotheses:

Main effects of species:

Null: All main effects are zero. In other words, the mean count does not differ between species, when averaging over (i.e., ignoring) seasons.

Saying: The mean count between species is the same is also correct. This phrasing applies to the rest as well.

Alternative: At least one main effect is not zero. The mean count between species does differ, irrespective of the season.

Main effects of season:

Null: All main effects are zero. In other words, the mean count does not differ between seasons when averaging over (i.e., ignoring) species.

Alternative: At least one main effect is not zero. The mean count between seasons does differ, irrespective of the species.

Remember the main effects are always averaged over the levels of the other factor and relative to the overall. Implicit in the statements above is that the difference in mean counts is relative to the overall mean.

Interaction effects:

Null: Species and season do not interact. The difference in species counts is consistent across seasons — i.e., the effect of species is the same regardless of the season.

Alternative: Species and season do interact. The effect of species depends on the season — i.e., the pattern of species counts changes across seasons.

12. Conduct the ANOVA hypothesis test by hand, present the results in a table. Use a 5% significance value.

$$SS_{\{Total\}} = \sum_{\{i,j,k\}} (Y_{ijk} - \bar{Y}_{...})^2 = 1305.9583$$

Sums of squares for main effects of Species (Notice that we can just sum the squared main effects):

$$SSA = b \cdot r \cdot \sum_{\{i=1\}}^{\{a\}} (\bar{Y}_{i..} - \bar{Y}_{...})^2$$

$$= 4 \cdot 3 \cdot [(-4.4583)^2 + (4.4583)^2] = 477.0417$$

Sums of squares for main effects Season (Notice that we can just sum the squared main effects)

$$SSB = a \cdot r \cdot \sum_{\{j=1\}}^{\{b\}} (\bar{Y}_{.j.} - \bar{Y}_{...})^2$$

$$= 2 \cdot 3 \cdot [(-1.625)^2 + (-0.625)^2 + 0.5417^2 + 1.7083^2] = 37.4583$$

Sums of squares due to the interaction: (Notice that we can just sum the squared interaction effects)

$$SSAB = r \cdot \sum_{\{i=1\}}^{\{a\}} \sum_{\{j=1\}}^{\{b\}} [(AD)_{\{ij\}}]^2$$

$$= 3 \cdot [(7.625)^2 + (-7.625)^2 + (2.2917)^2 + (-2.2917)^2 + (-2.875)^2 + (2.875)^2 + (-7.0417)^2 + (7.0417)^2] = 727.4583$$

Sums of squares error

$$SSE = SST - SSA - SSB - SSAB = 1305.9583 - 477.0417 - 37.4583 - 727.4583 = 64$$

Source	DF	SS	MS	F
Species (A)	a - 1 = 2 - 1 = 1	477.0417	477.0417 / 1 = 477.0417	477.0417 / 4 = 119.26
Season (B)	b - 1 = 4 - 1 = 3	37.4583	37.4583 / 3 = 12.4861	12.4861 / 4 = 3.12
Interaction (A×B)	(a-1)(b-1) = 1 x 3 = 3	727.4583	727.4583 / 3 = 242.4861	242.4861 / 4 = 60.62
Error	ab(n-1) = 4x4x(3-1) = 16	64	64 / 16 = 4	--
Total	N - 1 = 24 - 1 = 23	1305.9583		

13. How would you interpret each of the mean squares?

MSA (Species) was 477.0417, this is the estimated variation in insect counts due to differences between the two species, averaged across all seasons. There is very large variation in insect counts between species.

MSB (Season) was 12.4861, this is the estimated variation in insect counts due to differences between seasons, averaged across the two species. There is a smaller amount variation across the four seasons.

MSAB (Interaction) is the estimated variation due to the interaction between species and season. In other words, how much the effect of season depends on the species (and vice versa). The large MS suggests that the seasonal effect differs strongly between the two species (and vice versa)

14. What conclusion can you draw from the table about the effects? Calculate p-values using tables and R.

F-test for main effects of Species

F test statistic: 119.26

Null distribution is F with degrees of freedom 1 and 16.

Using the tables we can find

Critical value for 5% of 4.49, so since our test statistic is much larger than this we have strong evidence against the null hypothesis of no main effects. We conclude that at least one main effect is not zero. In this case we have only two means, so that suggests that the mean counts of species was different whilst ignoring the season.

We can get an approximate p-value from the tables: The largest value is 8.53 with a area of 0.01 to the right. So that means our p-value is smaller than 0.01 because the test statistic is greater than that critical value.

In R: `pf(119.26, df1 = 1, df2 = 16, lower.tail = FALSE)`

F-test for main effects of Species

F test statistic: 3.12

Null distribution is F with degrees of freedom 3 and 16.

Critical value at 5%: 3.24

F test stat < F critical value. No evidence against the null hypothesis, all main effects are zero.

For the p-value, notice that the test statistic falls between 2.46 and 3.24. The first critical value has an area to right of 0.1 and the second an area of 0.05. Our test statistic falls between, which means the p-value is between those two areas. So $0.05 < p\text{-value} < 0.1$!

In R: `pf(3.12, df1 = 3, df2 = 16, lower.tail = FALSE)`

F-test for interaction effects

F test statistic: 60.62

Null distribution is F with degrees of freedom 3 and 16.

Critical value at 5%: 3.24

F test stat >> F critical value. Evidence against the null hypothesis, at least one interaction term is not zero. The factors do interact. The effect of species on counts (so differences between mean counts) is different depending on the season.

In R: `pf(60.62, df1 = 3, df2 = 16, lower.tail = FALSE)`

15. Fit the model in R, extract the parameter estimates and obtain the ANOVA table. The data is in a CSV file called “stick_insects.csv”.

```
mod <- aov(Count ~ Season + Species + Season:Species, data = stick_insects)
model.tables(mod, type = "means")
model.tables(mod, type = "effects")
summary(mod)
```

Part 2

Complete an R analysis with the data below:

Here we are concerned with possible differences in yield among types of seed, among types of fertilizer, and the possibility that the two factors may interact. The data is in a CSV file called “maize_yield.csv”.

```
# =====
```

```
# MAIZE YIELD: Analysis
```

```
# =====
```

```
# Load and shuffle stick insect data
```

```
maize_yield <- read.csv("maize_yield.csv") # file path should be where you have stored the  
data set, i.e. in your working directory, or folder in your working directory
```

```
# EDA
```

```
summary(maize_yield)
```

```
maize_yield$SeedType <- as.factor(maize_yield$SeedType)
```

```
maize_yield$Fertilizer <- as.factor(maize_yield$Fertilizer)
```

```
# -----
```

```
# Assumption Checks
```

```
# -----
```

```
# Boxplot and stripchart
```

```
boxplot(Yield ~ SeedType + Fertilizer, data = maize_yield,
```

```
      main = "Boxplot of Maize yield by Treatment")
```

```

stripchart(Yield ~ SeedType + Fertilizer, data = maize_yield,
           vertical = TRUE, add = TRUE, method = "jitter", pch = 20, col = "blue")

# Standard deviations by treatment group
sds <- tapply(maize_yield$Yield, list(maize_yield$SeedType, maize_yield$Fertilizer), sd)
max(sds) / min(sds)

# Q-Q plots for selected treatment combinations,
par(mfrow = c(2, 2)) # Arrange plots in 2x2 grid

qqnorm(subset(maize_yield, SeedType == "A-402" & Fertilizer == "Fert II")$Yield,
       main = "Q-Q: Spring, Megacrania")
qqline(subset(maize_yield, SeedType == "A-402" & Fertilizer == "Fert II")$Yield)

qqnorm(subset(maize_yield, SeedType == "B-894" & Fertilizer == "Fert IV")$Yield,
       main = "Q-Q: B-894, Fert IV")
qqline(subset(maize_yield, SeedType == "B-894" & Fertilizer == "Fert IV")$Yield)

qqnorm(subset(maize_yield, SeedType == "C-952" & Fertilizer == "Fert III")$Yield,
       main = "Q-Q: C-952, Fert III")
qqline(subset(maize_yield, SeedType == "C-952" & Fertilizer == "Fert III")$Yield)

qqnorm(subset(maize_yield, SeedType == "C-952" & Fertilizer == "Fert V")$Yield,

```

```
    main = "Q-Q: C-952, Fert V")  
qqline(subset(maize_yield, SeedType == "C-952" & Fertilizer == "Fert V")$Yield)
```

```
par(mfrow = c(1, 1)) # Reset plotting layout
```

```
# Dotchart of Yields
```

```
dotchart(maize_yield$Yield,  
         labels = "",  
         main = "Dotchart of Stick Insect Yields",  
         cex = 0.7)
```

```
# -----
```

```
# Calculate Means and Effects
```

```
# -----
```

```
# Treatment means
```

```
(treatment_means <- tapply(maize_yield$Yield, list(maize_yield$SeedType,  
maize_yield$Fertilizer), mean))
```

```
# Marginal means
```

```
(Fertilizer_means <- tapply(maize_yield$Yield, maize_yield$Fertilizer, mean))
```

```
(SeedType_means <- tapply(maize_yield$Yield, maize_yield$SeedType, mean))
```

```
# Grand mean
```

```
(grand_mean <- mean(maize_yield$Yield))
```

```

# Effects (deviation from grand mean)

(Fertilizer_effects <- Fertilizer_means - grand_mean) # Should sum to zero

(SeedType_effects <- SeedType_means - grand_mean)


# -----

# Fit Two-Way ANOVA Model

# -----


mod <- aov(Yield ~ SeedType + Fertilizer + SeedType:Fertilizer, data = maize_yield)


# Estimated means and effects

model.tables(mod, type = "means", se = TRUE)

model.tables(mod, type = "effects", se = TRUE)


# ANOVA table

summary(mod)


# -----

# Interaction Plot

# -----


interaction.plot(maize_yield$SeedType, maize_yield$Fertilizer, maize_yield$Yield,

```

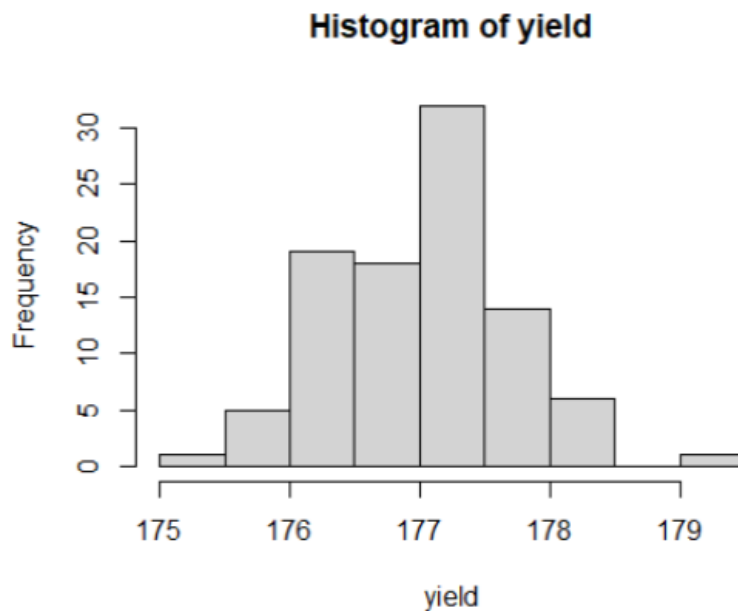
```
col = c("red", "blue", "green", "orange", "pink"), pch = c(1, 2), lty = c(1, 2),

main = "Interaction Plot: Yield by SeedType and Fertilizer")
```

Part 3

1. You are researching which type of fertilizer and planting density produces the greatest crop yield in a field experiment. You assign different plots in a field to a combination of fertilizer type (1, 2, or 3) and planting density (1 = low density, 2 = high density), and measure the final crop yield in bushels per acre at harvest time. To start off you perform a two-way ANOVA to test if there are significant differences in crop yield means for the given factors.

Source	Df	Sum Sq	Mean Sq	F value
fertilizer	A	6.07	3.03	D
density	B	5.12	5.12	15.19
fertilizer:density	C	E	0.21	0.63
Residuals	90	30.34	0.34	



- a) Determine the values of A, B, C, D, and E.

Fertilizer has 3 levels $\Rightarrow df_A = 3 - 1 = 2 \Rightarrow \mathbf{A = 2}$

Density has 2 levels $\Rightarrow df_B = 2 - 1 = 1 \Rightarrow \mathbf{B = 1}$

Interaction df $\Rightarrow (3 - 1)(2 - 1) = 2 \cdot 1 = 2 \Rightarrow \mathbf{C = 2}$

Interaction SS: $SS_{AB} = MS_{AB} \times df_{AB} = 0.21 \times 2 = 0.42 \Rightarrow \mathbf{E = 0.42}$

Fertilizer F-value: $F = \frac{MS_A}{MS_E} = \frac{3.03}{0.34} = 8.91(\text{approx}) \Rightarrow \mathbf{D \approx 8.91}$

- b) State the hypotheses associated with the third line in the ANOVA output. What conclusion is reached?

Null: There is no interaction between fertilizer and density. The effect of fertilizer on mean crop yield does not depend on the level of planting density (and vice versa).

Alt: There is an interaction between fertilizer and density. The effect of fertilizer on mean crop yield depends on the level of planting density (and vice versa).

- c) Comment on the ANOVA assumptions.

(1) The histogram of yield appears approximately symmetric and bell-shaped, with no strong skewness or outliers. This suggests that the normality assumption for the residuals is reasonable.

(2) Based on the information, it is evident that the samples are independent (different plots are used). This assumption is reasonable if plots were randomly assigned to the fertilizer–density combinations and that measurements on different plots do not influence one another. This is a design-based assumption and cannot be assessed directly from the data shown here.

(3) cannot comment on constant variance.

- d) How many observations were used in this experiment?

Residual df = 90. In a balanced 3×2 factorial with n replicates per cell:

$$df_E = ab(n - 1) = 6(n - 1) = 90 \Rightarrow n - 1 = 15 \Rightarrow n = 16$$
$$N = abn = 6 \times 16 = 96$$

2. You have recently started a new business and are interested in advertising in your local newspaper. You find data that gives you the number of inquiries resulting from an advertisement. The data also provides information on which day of the week (Monday to Friday) and section of the paper (news, business, or sports) the advertisement was posted. You perform a two-way ANOVA to investigate.

ANOVA output

Source	df	Sum Sq	MS	F	Pr(>F)
Section	2	53.73	26.87	15.30	A
Day	4	146.83	36.71	20.91	B
Section:Day	8	135.77	16.97	9.67	C
Residuals	45	79.00	1.76		

Is there reason to believe that you should consider in which section and on which day you should post your advertisement? You should mention the different variables considered, and associated statistics.

`A = pf(15.30,2,45,lower.tail = FALSE)`

`[1] 8.521993e-06`

`B = pf(20.91,4,45,lower.tail = FALSE)`

`[1] 8.516429e-10`

`C = pf(9.67,8,45,lower.tail = FALSE)`

`[1] 1.119985e-07`

Interpretation / conclusion

At $\alpha = 0.01$, all three tests are significant (very small p-values).

- Section effect: inquiries differ by section.
- Day effect: inquiries differ by day.
- Interaction: the effect of section depends on the day (and vice versa).

Yes, you should consider both section and day, especially because the interaction is significant. The best section cannot be chosen without considering which day it runs.

3. List all the ANOVA assumptions and suggest a way of checking if each of the assumptions holds true.

Independence: ensured by randomisation; check design and residual patterns (time/space) by inspecting a dot chart.

Normality of errors: QQ-plot/histogram of residuals

Equal variances: residuals vs fitted; boxplot of response by treatment

No severe outliers: boxplot of response by treatment

4. You are researching if task performed on a computer (point-select, drag-select, etc.) and/or the device used (mouse, stylus, etc.) affects the time it takes to complete a task. The data observed is shown below.

	Device 1	Device 2	Device 3
Task 1	11, 10, 10	15, 17, 13	20, 11, 20
Task 2	18, 14, 23	13, 15, 20	14, 13, 16

At the 10% significance level test whether there exists a significant interaction between Task performed and Device used.

	Df	Sum Sq	Mean Sq	F value
device	2	6.33	3.17	0.28
task	1	20.06	20.06	1.80
device:task	2	88.11	44.06	3.95
Residuals	12	134.00	11.17	

Test statistic for interaction:

$$F_{AB} = 3.945 \text{ with } (2,12) \text{ df}$$

Critical value:

$$F_{2,12}(0.10) \approx 2.807$$

Since $3.945 > 2.807$, reject H_0 at the 10% level.

(Equivalently, p-value ≈ 0.048 , which is < 0.10 .)

Conclusion: There is significant evidence of an interaction between Task and Device at the 10% significance level.

Interpretation (what does the interaction mean in context?)

The effect of device depends on the task:

- For Task 1, Device 1 is fastest (mean 10.33), Device 3 is slowest (mean 17.00).
- For Task 2, Device 1 is slowest (mean 18.33), and Device 3 is fastest (mean 14.33).

So you cannot recommend a single “best device” without specifying the task: performance rankings change across tasks (non-parallel lines in an interaction plot).

5. In a study on the effect of advertising, TV ads were shown to a group of twenty subjects. Afterwards their attitude to the advertised product was measured. Of the twenty, ten had been classified by a psychologist as having an extrovert personality, while the other ten had an introvert personality. Five subjects from each personality group were randomly chosen to hear the advertisement at high volume and the other five heard it at normal volume.

- a) Calculate as many as possible of the missing values in the ANOVA table below.

Source	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Personality	1				
Volume					4.61
Interaction			15		11.39
Residuals					

Source of Variation	DF	SS	MS	F
Personality	1			
Volume	1	69.15	69.15	4.61
Interaction	1	170.85	170.85	11.39
Error	16	240	15	
Total	19			

- b) Is there evidence that the mean for high volume differs from that for normal volume? Give a p-value.

There is weak evidence for a difference in means ($F = 4.61$, $p\text{-value} = 0.047$) at the 5% significance level.

```
pf(4.61,1,16,lower.tail = F)
[1] 0.0474571
```

- c) Do the factors Personality and Volume interact? Test using a p-value approach.

There is strong evidence for an interaction between personality type and volume on attitude towards the advertised product ($F = 11.39$, $p\text{-value} = 0.004$) at the 5% significance level.

```
pf(11.39,1,16,lower.tail = F)
[1] 0.003859153
```

d) Given the following means:

	High volume	Low volume
Extrovert	10.6	7
Introvert	7.2	6

Draw a sketch to show the nature of the interaction, if any. Then, interpret this in words.

The sketch should show two clearly non-parallel lines (it does not matter which variable you choose for the x-axis). The difference between high volume and low volume should be greater for extroverts. The effect of volume on the attitude towards the advert depends on whether the person is introverted or extroverted (and vice versa). Specifically, the effect of volume seems to be greater for extroverts compared to introverts.

6. In a medical study, the growth of bacteria is thought to be affected by temperature and pH. The experimental data below are of bacterial growth at four different temperatures and three different pH levels. There are two replications. The design is thus a 4×3 factorial design with two replications.

Temperature (°C)	pH 5.5	pH 6.5	pH 7.5
25 rep 1	9	18	36
25 rep 2	11	20	44
30 rep 1	13	23	27
30 rep 2	17	27	33
35 rep 1	18	27	23
35 rep 2	22	33	27
40 rep 1	22	20	7
40 rep 2	28	24	13

Given that the sums of squares for the interaction is equal to 1203, carry out an ANOVA to investigate the effects of temperature and pH on bacterial growth. Is there evidence that the two factors interact?

Given $SS_{AB} = 1203$.

df

Temp: $4 - 1 = 3$

pH: $3 - 1 = 2$

Interaction: $(3)(2) = 6$

Error: $ab(r - 1) = 12(1) = 12$

Total: $N - 1 = 24 - 1 = 23$

Sums of squares (from the data)

$SS_T = 1797.833$

$SS_{\text{Temp}} = 116.500$

$SS_{\text{pH}} = 330.333$

$SS_{\text{Int}} = 1203.000(\text{given})$

$SS_E = 148.000$

ANOVA table

Source	df	SS	MS	F	p-value
Temperature	3	116.500	38.833	3.149	0.0648
pH	2	330.333	165.167	13.392	0.000877
Temp × pH	6	1203.000	200.500	16.257	3.97×10^{-5}
Error	12	148.000	12.333		
Total	23	1797.833			

Conclusion ($\alpha = 0.05$)

Interaction: significant ($p < 0.05$) $\Rightarrow$ yes, the factors interact. Evidence of strong interaction between temperature and pH. The effect of pH on bacterial growth depends on the level of temperature (and vice versa).

7. *“If the investigator confines his attention to any single factor we may infer either that he is the unfortunate victim of a doctrinaire theory as to how experimentation should proceed, or that the time, material or equipment at his disposal is too limited to allow him to give attention to more than one aspect of his problem...”*

This is a quote from R. A. Fisher. What is he referring to?

Fisher is referring to the importance and efficiency of factorial experimentation: studying more than one factor at a time (and their interaction), rather than “one-factor-at-a-time” experiments. A single-factor focus is either:

- a misguided belief about experimentation, or
- forced by limitations (time/material/equipment).

He is advocating factorial designs to learn about main effects and interactions efficiently.

8. Here are the results of a factorial study involving two treatment factors A and B. Given that SS Interaction is equal to 80, carry out an Analysis of Variance and give clear conclusions to the three hypothesis tests performed.’

	B1	B2	B3
A1	5	9	3
	3	9	8
	3	13	3
	8	6	3
	6	8	3
A2	0	0	0
	2	0	3

	0	0	7
	0	5	5
	3	0	5

A has 2 levels, B has 3 levels, 5 reps per cell.

df

$$df_A = 1$$

$$df_B = 2$$

$$df_{AB} = 2$$

$$df_E = ab(n - 1) = 6 \cdot 4 = 24$$

$$df_T = 29$$

From the data (and consistent with given $SS_{AB} = 80$):

$$SS_A = 120$$

$$SS_B = 20$$

$$SS_{AB} = 80$$

$$SS_E = 120$$

$$SS_T = 340$$

ANOVA table

Source	df	SS	MS	F	p-value
Factor A	1	120	120	24.0	5.37×10^{-5}
Factor B	2	20	10	2.0	0.157
A×B	2	80	40	8.0	0.00218
Error	24	120	5		
Total	29	340			

Conclusions ($\alpha = 0.05$)

- Factor A: significant ($p < 0.05$) $\Rightarrow$ A affects the mean response.
- Factor B: not significant ($p = 0.157$) $\Rightarrow$ no evidence B affects the mean overall.
- Interaction: significant ($p = 0.00218$) $\Rightarrow$ effect of B depends on A (or vice versa).

There is a significant A×B interaction; interpret main effects with caution.